

Scanning-projection based stereolithography: Method and structure



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ABSTRACT

Projection-based stereolithography (PSL) is an efficient way for rapid fabrication of 3D structures. However, it is suitable for small parts requiring high resolution. The scanning-based SL (SSL) covers a larger area with a lower resolution. Fabrication time in PSL is intrinsically less than SSL due to exposing an entire image field in a single shot. This study introduces a new scanning-projection based stereolithography (SPSL) to include the advantages of both former methods. SPSL takes advantage of a digital micromirror device (DMD) and a moving stage to combine the projection and scanning methods. The DMD moves continuously over the medium while the projected pattern is continuously updated to allow the fabrication of large layers.

After modeling the approach, an experimental setup is assembled to produce real parts. An economical ultra violet light-emitting diode (UV-LED) is used as a light source for curing. Seven 3D models with defined geometry consisted of circular, rectangular, complex and overhanging parts are selected. Five copies of each model are built. The experimental results show that the method is feasible and versatile. The achieved standard deviation is lower than 32 μm .

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1. Introduction

Rapid prototyping (RP) reduces the time between product conception and final design to fabricate complex shapes. Stereolithography (SL) is one of the most important methods in RP. There are two different SL techniques distinguished by their patterning process:

- Scanning-based SL (SSL) or direct writing (Fig. 1a).
- Projection-based SL (PSL) that is also called maskless or dynamic mask method (Fig. 1b).

There are distinct advantages and disadvantages associated with each technology in terms of resolution, layer thickness, accuracy, cost efficiency, and throughput. In SSL a focused laser beam with small diameter (~ 0.1 mm) scans large areas by small rotations of a XY Galvo-mirror [1].

PSL uses a spatial light modulator (SLM) as dynamic masks for generating 2D patterns with μ -scale resolution. SLM is an optical module selectively turns light pixels/strips on or off. LCD, liquid

crystal on silicon (LCoS) and digital micromirror device (DMD) are the most common SLMs used in recent works as dynamic mask generators [3–5,7–12]. LCD is transmissive while LCoS and DMD are reflective SLMs.

SLM application for dynamic mask projection in SL was first introduced by Bertsch co-workers [3,4]. They built an SL machine with LCD to generate dynamic masks for micro-stereo-photolithography. Monochromatic light (argon laser) as an advantage and inhomogeneity of laser beam as the disadvantage was addressed in their work.

LCD has a few intrinsic drawbacks such as large pixel size, low filling ratio and low transmittance (specially for shorter wavelength) which reduces the efficiency of light illumination [5].

To overcome the low performance of LCDs, Sun et al. used DMD for its high contrast and resolution [5]. Mercury I-line emission was chosen as the illumination light source.

The transmittance observed by typical DMDs in visible region is approximately 97% (single-pass through two surface transitions). But the transmittance of UV spectrum (355–400 nm) is dramatically reduced to 60% [6]. Recently, a PSL system is reported based on a high resolution LCoS chip [7]. Nevertheless, the performance of the system is not compared with other SLMs.

PSL has not only micro-resolution but also a higher fabrication speed in compare to SSL, due to single shot exposure of the whole

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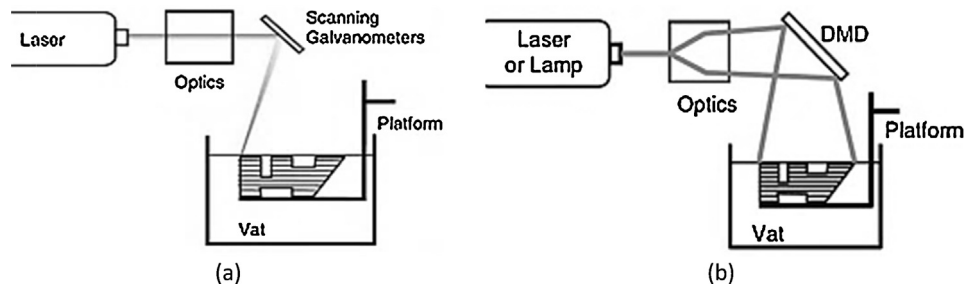


Fig. 1. Schematic of (a) SSL and (b) PSL [2].

pattern [5,8]. The key challenge in PSL is its limited resolution and small array size of DMDs. Projecting a pattern with a DMD array over a larger part/platform surface will cause a larger pixel size and as a result a lower resolution.

The highest available resolution of DMD is 1920×1080 micro-mirrors [9]. With this DMD, the minimum feature size in each layer for a micro platform ($9.6 \text{ mm} \times 5.4 \text{ mm}$) is $5 \mu\text{m} \times 5 \mu\text{m}$. But, it results in the minimum feature size of $200 \mu\text{m} \times 200 \mu\text{m}$ in a macro platform ($384 \text{ mm} \times 216 \text{ mm}$) due to magnification of the pattern.

To overcome the limitations of projection method (creating a large part with high resolution), three approaches exist. The first one is coupling of multiple DMDs to make a larger array. This is very expensive and it is difficult to obtain uniform illumination.

The second method is a moving DMD that stops over a portion of platform, exposing the area underneath, moving to the next position and repeating this cycle. This method is usually called “step-and-repeat” and may cause increased polymerization in overlaps. This process is demonstrated by Ha [10] and Zheng [7] for batch production and Micro Stereolithography.

The third method is the subject of this work and it is not reported before. It is called scanning-projection stereolithography (SPSL). It combines the advantages of both SSL and PSL. In this method, DMD moves continuously over the area of the medium while the projected image is continuously updated. In SPSL scanning and projection are carried simultaneously while in step-and-repeat projection and movement are done separately; one after the other. Thus SPSL should clearly be faster. Also, it is anticipated that the exposure non-uniformity, which is frequently reported

[11,12], may be improved; because the illumination cumulatively received by each point from different μ -mirrors during the scan.

In this investigation, a SPSL system is modeled and built. Several parts are manufactured to prove the feasibility of the idea. An economical high-power UV light-emitting diode (UV-LED) is innovatively used as a light source for curing as well.

2. Process description and background

Fig. 2 shows the schematic of SPSL method. It is composed of a scanning stage, a resin bath, a UV projector and a Z-stage. The scanning stage moves along X and Y directions over the resin bath. UV light is emitted toward DMD and is reflected through the lens over the resin surface. A computer, controls the reflectivity of each DMD μ -mirror. In this way, a selective exposure area is created.

The scanning stage moves continuously while the computer changes the reflection status of each μ -mirror to create a dynamic exposure pattern. Once the exposure of a layer is finalized, the Z-stage moves and allows the liquid resin to form a new layer to be exposed.

2.1. Top and bottom exposure

The structure described above can be implemented in two distinct ways. Based on construction and irradiance direction, these methods are top exposure (TE) and bottom exposure (BE). In TE, the top surface of resin is exposed to a 2D pattern while the platforms

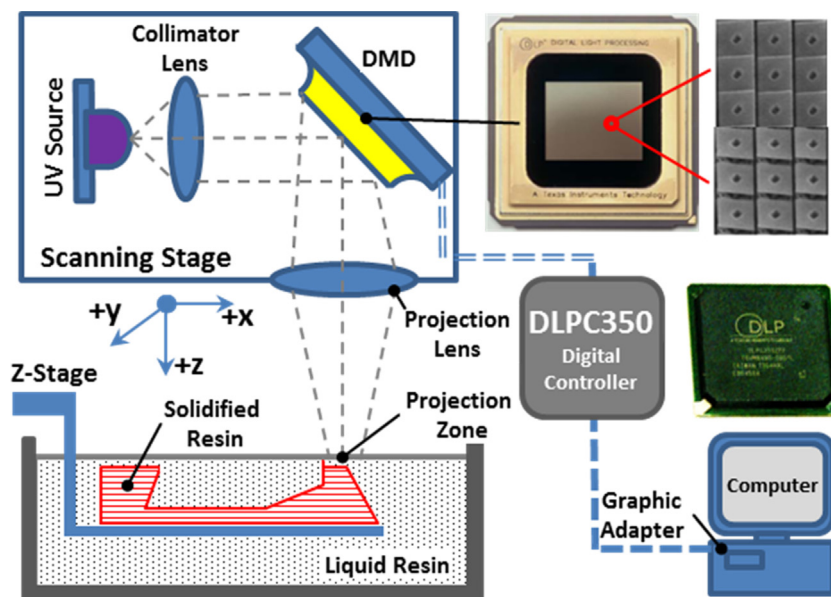


Fig. 2. Schematic drawing of SPSL.

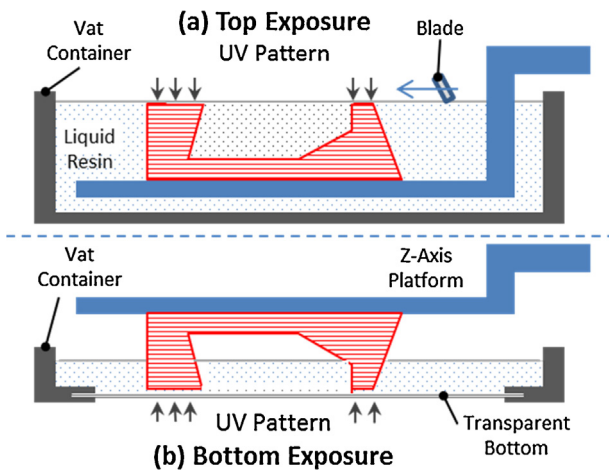


Fig. 3. (a) Top exposure vs. (b) bottom exposure.

dips down for each new layer (Fig. 3a). In BE patterns are illuminated on the bottom surface of a transparent container (Fig. 3b) and platform is lifted up to generate a fresh layer.

This presents several advantages and some disadvantages in compared to TE. From an economical point of view, it consumes significantly less resin, since the whole fabricated solid is not required to be immersed inside the resin. The resin surface settling time and particularly re-coater blade is not required in BE. Therefore, it is more convenient for micro-stereolithography.

Because each new layer is formed at the bottom of the resin, the layers are completely flat in compare to TE. Consequently, a better control of the thickness of every fresh layer is achievable in BE [13]. In addition, in TE, oxygen inhibition occurs which is not the case for BE where the process layer is not exposed to air.

Furthermore, in BE, there is a possibility for the polymerized resin to stick to the transparent window. This may cause layer damage during the platform lift up. Also in fabrication of a large part, the weight of the part is sometimes a limit. Heavy parts may separate from the platform or break in weak sections. This problem could be solved by adding supports in weak sections. In this work, a SPSL

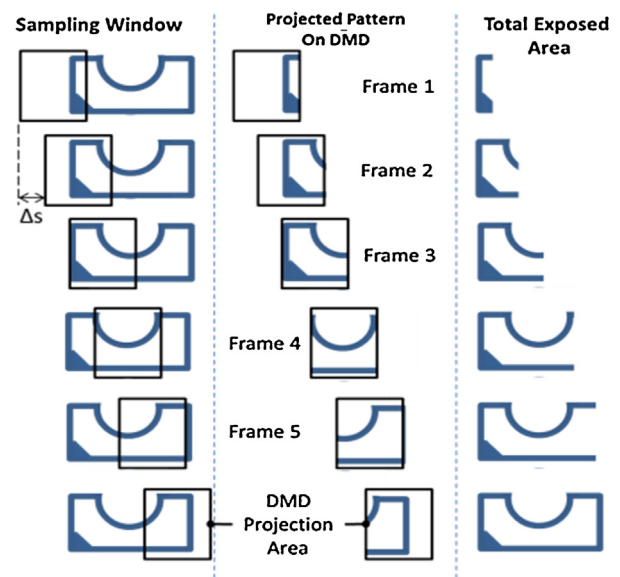


Fig. 4. Window sampling used in the discussed method.

system is implemented based on BE technique and polyethylene terephthalate (PET) sheet is used as UV-vis transparent window.

2.2. Window sampling

As mentioned earlier in Section 1, when the whole pattern does not fit into the projection area, window sampling of pattern is used which is very similar to video animation frames (Fig. 4). In each sampling period, the projected part of the pattern is defined as a single frame. The displacement of sampling window which is corresponding to the displacement of scanning head in each sampling period is Δs . While sampling window moves, a frame evolves as showed in Fig. 4.

The best quality is achieved when the image shift in DMD array is one pixel column (minimum binary value) at each sampling time. Therefore, the difference between two consequence frames is 1 pixel column (Fig. 5). If pixel pitch in the image plane (resin

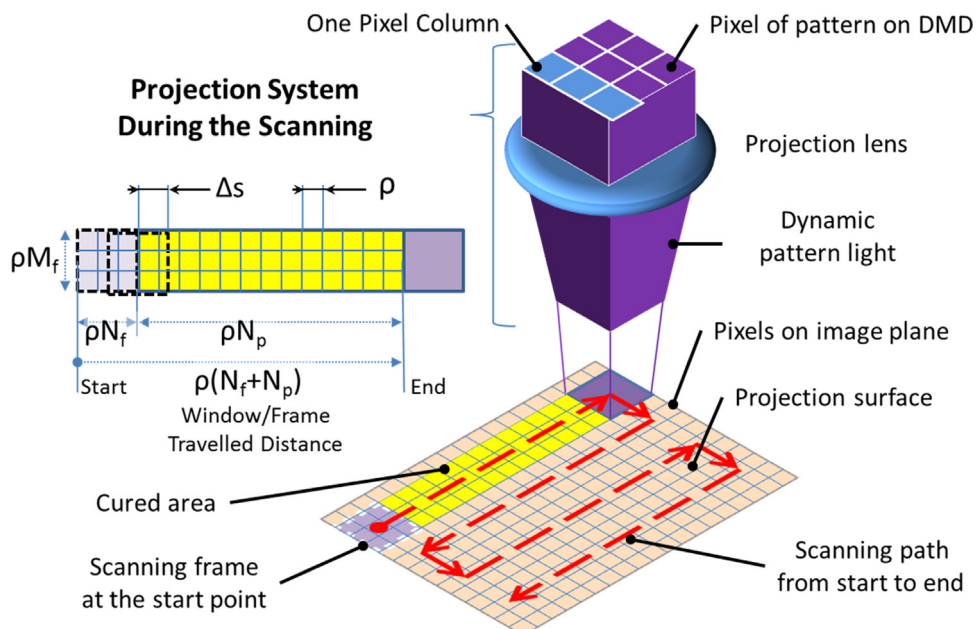


Fig. 5. Schematic design of projection system while scanning a pattern.

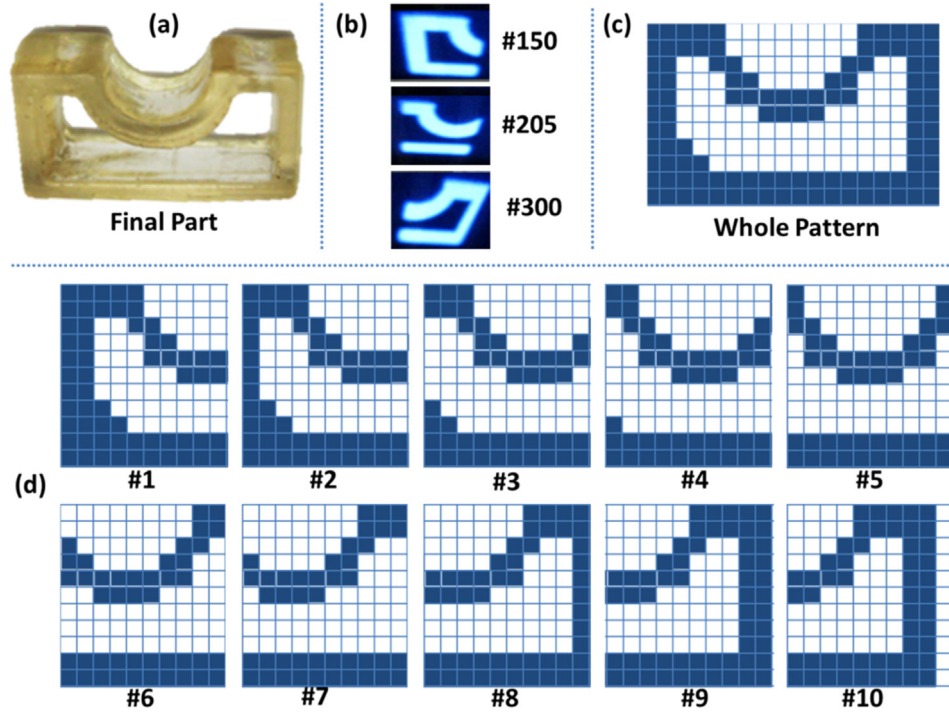


Fig. 6. A part generated by SPSL and the evolution of reflected patterns on DMD. (a) Real part, (b) real images of the patterns on DMD, (c) schematic of the whole pattern (18 × 10 pixels), and (d) Schematic of consecutive frames generated on DMD during the scanning process.

surface) is defined by ρ , to avoid image elongation Δs must be equal to ρ . The time it takes for such a displacement is called single pixel shift time (T_{SPS}). T_{SPS} cannot be less than $1/F_{max}$, where F_{max} is the maximum frame rate of the DMD.

The schematic of SPSL patterning is illustrated in Fig. 5. The frame size is $M_f \times N_f$ pixels. The total displacement of the frame is $\rho(N_f + N_p)$, where N_f and N_p are scanning frame width and exposure length respectively both in pixels unit (Fig. 5). The exposure height is equal to the frame height (ρM_f). The total number of frames generated for each layer is $N = N_f + N_p$.

Fig. 6 shows a part generated by SPSL and the evolution of reflected patterns on DMD. Three real high-resolution images of the illuminated pattern are captured by a camera and showed in Fig. 6b. Consecutive generated frames numbered from 1 to 10, are sketched in Fig. 6d. Each frame is generated by a single pixel shift for each row as described in Section 2.1. At each motion step, the generated frames are updated on DMD. Generally the frame width (10 pixels in Fig. 6) is shorter than the pattern width (18 pixels in Fig. 6).

3. Process modeling

While the window sampling is performed, each pixel on resin surface receives several exposures from traveling DMD. If the scanning head is traveling too fast, the resin surface will not receive enough energy to polymerize. On the other hand, slow displacement of head will result in over exposure. Therefore, a mathematical model is required to calculate the total energy received by a pixel on resin surface based on scanning speed. This formulation is developed based on diffractive optics and Beer–Lambert law. Maximum cure depth (C_d) of a single spot Gaussian light is obtained from Eq. (1) called “working curve equation” [1].

$$C_d = D_p \ln \left(\frac{E_{max}}{E_c} \right) \quad (1)$$

where D_p and E_c are light penetration depth and effective critical exposure respectively. E_{max} is the maximum center line exposure energy of Gaussian spot light (Fig. 7). In fact, C_d is the cure depth corresponding to E_{max} based on Beer–Lambert law. Both D_p and E_c are resin parameters obtained via experimental procedure discussed in Section 4.

By definition exposure (E) is the integral of the irradiance (I) over time. It is simply defined by Eq. (2) for time-independent irradiance.

$$E(\text{mJ}/\text{cm}^2) = I(\text{mW}/\text{cm}^2) \cdot T(\text{s}) \quad (2)$$

where T is the exposure time. For a single pass of a mirror, the energy received by a pixel on resin surface (E , mJ/cm²), depends on the number of active μ -mirrors traversing over pixel (N_f), the

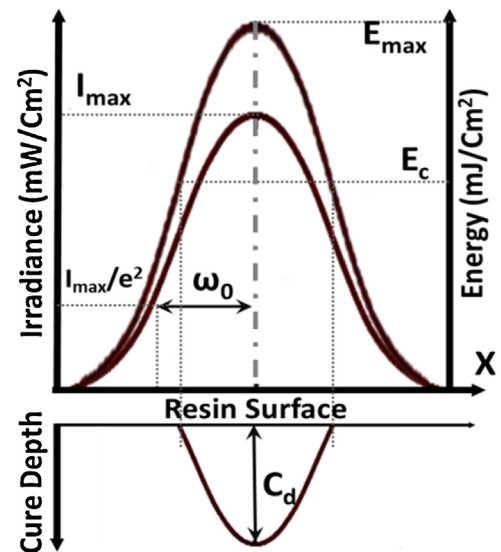


Fig. 7. The irradiance distribution of individual μ -mirror.

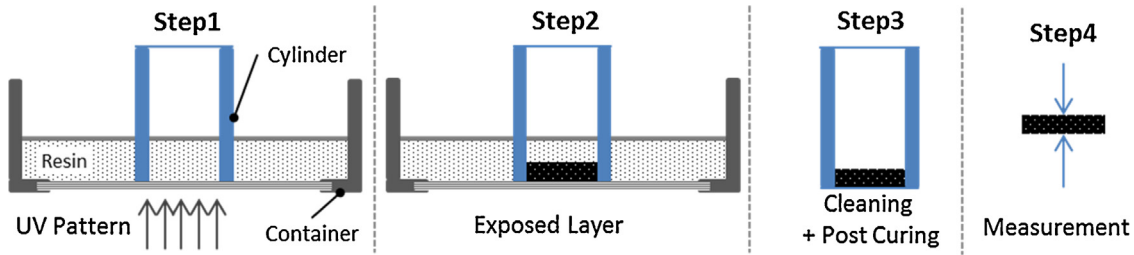


Fig. 8. Experiment on obtaining working curve.

irradiance reflected by each μ -mirror (I , mW/cm²), and T_{SPS} . This energy is obtained from Eq. (3):

$$E = I \cdot T_{SPS} \cdot N_f \quad (3)$$

If $\Delta s = \rho$ (Fig. 5) and the scanning speed is V_s , then $T_{SPS} = \rho/V_s$. Thus, Eq. (3) is rewritten as:

$$E = I \cdot \rho \cdot \frac{N_f}{V_s} \quad (4)$$

where I is assumed constant over time. In practice I may vary along DMD array and also on a pixel when frame is moving. In order to achieve the maximum energy reflected by a μ -mirror (E_{max} in Fig. 7), the maximum irradiance I_{max} is replaced in Eq. (4):

$$E_{max} = I_{max} \cdot \rho \cdot \frac{N_f}{V_s} \quad (5)$$

Eq. (5) implies that a bigger energy is received by resin if the stage moves slower. The goal, of course, is to have the fastest moving stage while projecting high enough irradiance. The minimum value of T_{SPS} is limited by both F_{max} and E_c . For most DMDs, F_{max} is between 10 kHz and 22 kHz which is very fast. Therefore, the limiting factor will be E_c .

By omitting E_{max} between Eqs. (1) and (5), the convenient value for V_s is obtained based on the required cure depth (C_d) as follows:

$$V_s = I_{max} \cdot \rho \cdot \frac{N_f}{E_c} \cdot \exp\left(-\frac{C_d}{D_p}\right) \quad (6)$$

As Eq. (6) describes, the scanning speed (V_s) depends on resin optical parameters (E_c and D_p) which are not accurately reported by manufacturers. E_c and D_p are obtained as discussed in Section 4.

4. Extraction of photopolymer parameters

To find optical parameters of resin, E_c and D_p , the working curve should be extracted. If E from Eq. (2) is replaced in working curve equation (Eq. (1)), Eq. (7) is obtained:

$$C_d = D_p \ln\left(\frac{I \cdot T}{E_c}\right) = A \ln(T) + B \quad (7)$$

where $A = D_p$; and $B = D_p \ln(I/E_c)$. Eq. (7) is the time representation of working depth. To extract the working curve, the cured depths at different exposure times are measured and plotted against $\ln(T)$. The slope of the curve is D_p . E_c is obtained from B (interception of working curve with the line $C_d = 0$). A four-step experiment is carried out to plot the working curve at 395 nm wavelength (Fig. 8). The procedure is defined as follows:

1. A 6 mm transparent PET cylinder is placed in a polymer container for easy handling of the cured layer (step 1, Fig. 8). This is because

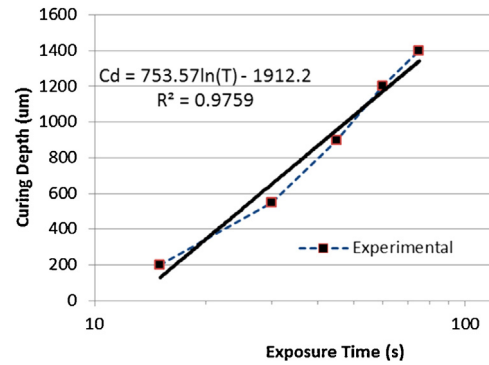


Fig. 9. Working curve of BEST PL-4151; experimental measurement (points); regression curve (solid line); semi logarithmic plot.

C_d is usually less than 1 mm. Therefore, it is difficult to handle the produced layer without a carrier (the PET cylinder).

2. Enough resin (less than 3 mm height) is poured into the container. This layer is fully exposed using a circular pattern (step 2, Fig. 8).
3. The PET cylinder with a solid layer is removed from the resin and is cleaned with Isopropyl Alcohol (IPA) for 5 min. The solid layer is then post cured for 4 min (step 3, Fig. 8).
4. The cured part thickness is measured using micro-meter (step 4, Fig. 8).

The layer thickness is a function of intensity which is controlled by the exposure time. This procedure is repeated for exposure times 15, 30, 45, 60 and 75 s (C_d between D_p and $4D_p$ as mentioned in [1]).

Several photo-polymers were evaluated for this research. “BEST PL-5141” resin from “BEST-Klebstoffe GmbH” was selected because of its low price, availability, its response wavelength (365–400 nm),¹ low viscosity (~ 80 cps at 30 °C) and transparency in both liquid/cured states. BEST PL-5141 is a single component, solvent-free and based on modified acrylic resin. However, it is not the best selection if tolerance and dimension are the priority.

The result of the curing depth versus exposure time is shown in Fig. 9. The dots and the solid lines show the experimental data and curve-fitted results, respectively. The results indicate that the curing depth (C_d) is almost linearly proportional to the natural logarithm of exposure time as Eq. (7) predicts. The value of regression correlation coefficient (R) is calculated by Excel and shown in the plot. The closer R is to ± 1 , the more linear the plot is. R must be greater than 0.99 to insure less than 1 mil error [1]. The experimental value of maximum irradiance is $I_{max} = 2.03$ mW/cm². From the experimental data (Fig. 9), $A = 754$ and $B = 1912$ are extracted. Therefore, $E_c = 25$ mJ/cm² and $D_p = 754$ μ m are obtained by Eq. (7).

The value of E_c is greater than conventional SL resins (4–8 mJ/cm² for acrylate and 10–20 mJ/cm² for epoxy based resins).

¹ Our DMD could reflect wavelengths larger than 380 nm.

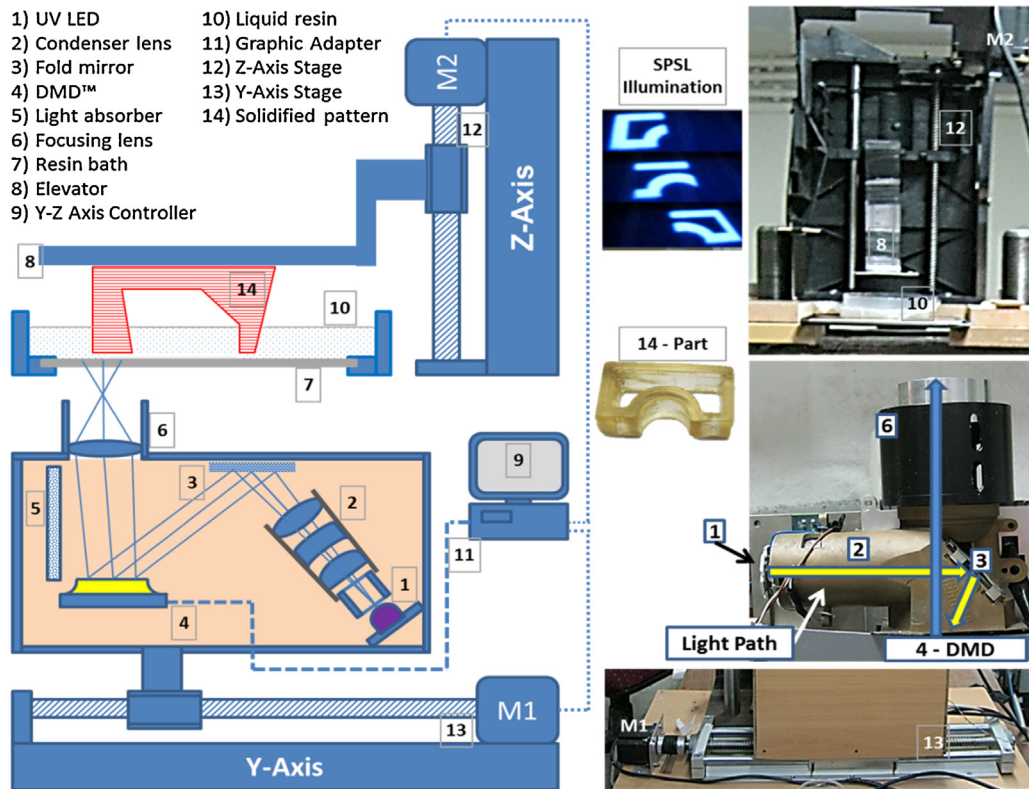


Fig. 10. Schematic and experimental setup of the SPSSL system.

This is because the reported value for E_c is for 365 nm wavelength (I-line). Here we use 395 nm (UV-vis) instead which requires more energy naturally.

5. Experimental setup

In order to study the feasibility of the SPSSL, a setup shown in Fig. 10 is prepared. An old DLP (digital light processor) projector is employed with some modifications to produce the UV light and generate the patterns. The projector lamp is replaced with UV LEDs.

When the machine is turned on, UV light (Fig. 10, #1) is transmitting through lens (Fig. 10, #2) in the modified DLP projector. DMD reflects the programmed pattern (Fig. 10, #4). UV light leaves DLP projector through the output lens (Fig. 10, #6) and passes through the transparent bath (Fig. 10, #7) to expose the resin at the bottom of the container. The bottom of polymer bath (Fig. 10, #7) is made from polyethylene terephthalate which is transparent to UV-vis and does not stick to the solidified polymer layer. All components are summarized in Table 1.

In this research, a high power UV LED is utilized to supply UV light source for PSL/SPSSL. It is controlled by a relay to turn on/off the UV source. Although some publications praise the high efficiency of UV LEDs but in fact they waste more than 75% of energy they receive [14]. The advantages of LEDs are their narrow wavelength, low price and higher switching rate. Table 2 shows a comparison between different UV sources. The lens system is designed to magnify the DMD pattern four times larger on resin surface. Silica lens was used since silica has good transmission in the UV region. To provide an accurate linear movement of the UV projector (Y-axis) and the building platform (Z-axis, Fig. 10, #8), two individual PC-controlled single axis stage is implemented (Fig. 10, Y and Z axes). The accuracy of layer thickness is determined by the Z-stage resolution. Typically, the resolution of the Z-stage should be much less than layer thickness. In order to ensure the resolution and

accuracy, slow movement is set for Z-axis movement. A MATLAB code is developed to create slicing patterns from STL file. Some of the important parameters such as building orientation, Layer thickness and structure support should be defined before slicing [15]. The generated slices are stored in a series of monochromatic bitmap files. Another MATLAB code is written to control the hardware consisting of loading the sliced patterns, transferring patterns to video interface and projection system, switching the UV light and controlling the motion of the stages.

Table 1

Components specifications and material properties used in experimental setup.

No.	Component	Description
1	UV LED	395 nm(± 5 nm) InGaN die material 1 W Power LED
2	Lens	Fused silica convex/concave lens from Thorlabs
3	Fold mirror	Aluminum coated metallic mirror from Thorlabs
4	DMD and DLP	Digital micromirror device and digital light processor from Texas Instrument, SVGA (800 $\times$ 600), $\pm 12^\circ$, 13.7 μ m pixel pitch, 10 kHz frame rate, from Infocus LP500
5	Light absorber	Anti-reflective light absorber
6	Objective lens	Fused silica convex lens from Thorlabs
7	Resin bath	Resin container with UV transparent bottom from thin polyethylene terephthalate (PET) sheet
8	Elevator	Home-made, refer to Z-axis stage (#13) for more details
9	PC-interface	Advantech PCI-1710 series, 16 bit multifunction card
10	Resin	BEST PL-5141 resin from BEST-Klebstoffe GmbH; cured tensile strength: 42 N/mm ² ; elongation at yield: 7%; density: 1.1 g/cm ³ at 23 $^\circ$ C
11	Graphic adapter	32 bit analog graphic adapter/PCI interface
12	Y-axis stage	Single axis, 10 μ m resolution, maximum speed 20 mm/s, drive 0.72 $^\circ$ stepper motor
13	Z-axis stage	Single axis, 4 μ m resolution, maximum speed 10 mm/s, drive 0.72 $^\circ$ stepper motor

Table 2
Comparison of UV sources [14].

Properties	LED	Lamp	Laser
Cost effective	Inexpensive	Moderate	Very expensive
Easy to use	Easy	Difficult	Moderate
Life time	Long (10,000 H)	Short (2000 H)	Short (2000 H)
Bandwidth	Narrow band	Broad band	Narrow band
Size	Compact	Large	Large
Intensity	Moderate	High	Low to moderate
Filter	No Need	Need	No need
Heat	Low	High	Moderate
Start-up time	Short (<10 ns)	Moderate	High
Oxon production	No	Yes	No
Mercury free	Yes	No	No
Cooling	Heat sink	Fan + heat sink	Fan + heat sink

6. Results and discussion

In order to show the feasibility of SPSL, seven different CAD models with different geometries were manufactured (Fig. 11). Four 3D

models with defined geometries consisted of circular and square patterns were considered. The parts are designed with repeated patterns to investigate the repeatability of the process. Each model is manufactured five times. Table 3 shows the manufacturing data for each model. An example showing how V_s is calculated is presented in Appendix.

In order to investigate the repeatability of the manufactured parts, two index dimensions (X and Y, Fig. 11) are measured for some of the models. The result is listed in Table 4.

The results of Table 4 for 20 tests show that the repeatability limit of SPSL under 2σ criteria is less than $64\ \mu\text{m}$ [16]. It should be noted again that the resin used here is not suitable for precise stereolithography. It is merely used as UV curable adhesive. Therefore, the topology might be disturbed at some points. As Fig. 11 shows, SPSL is quite feasible and can be extended to fabricate larger parts with acceptable repeatability.

To show the capability of SPSL to produce complex geometries, two different models consisted of an overhanging hook (Fig. 11k) and an industrial turbine blade (Fig. 11m) are fabricated.

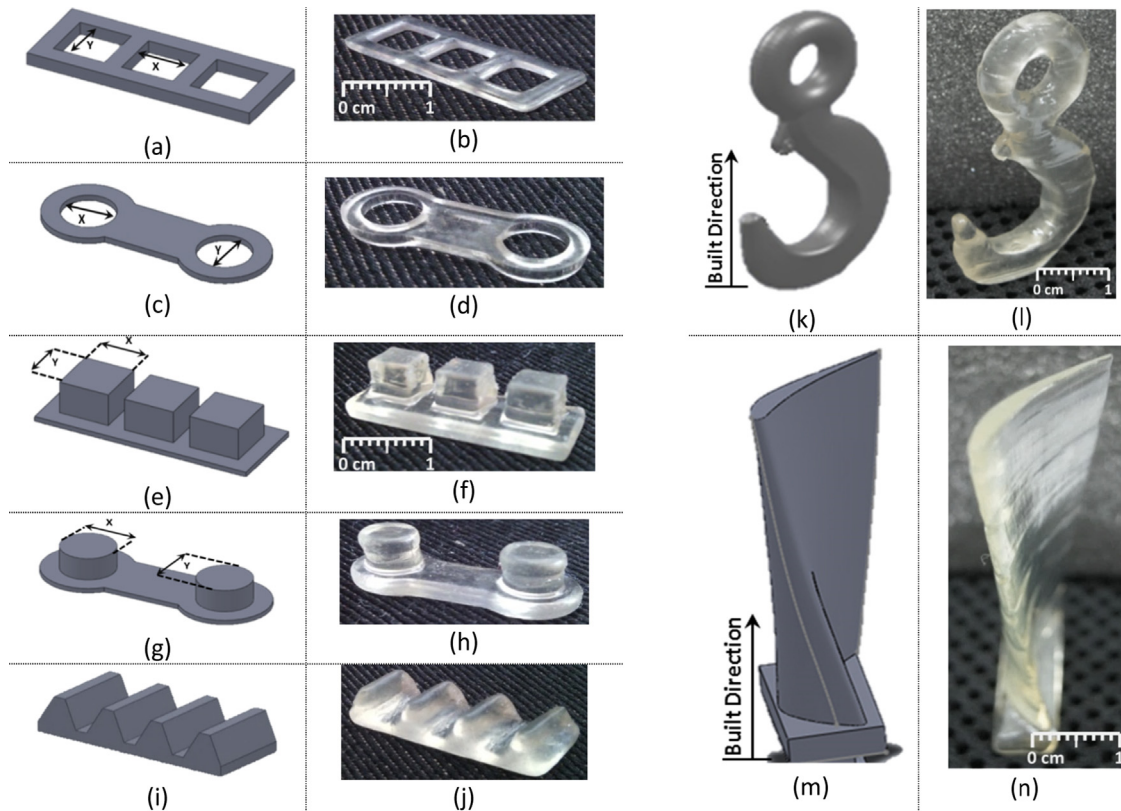


Fig. 11. CAD models and produced parts.

Table 3
Process parameters.

Model in Fig. 11	(a)	(c)	(e)	(g)	(i)	(k)	(m)
Total number of layer	50	150	50	150	65	400	462
Layer Thickness (μm)	40	40	40	40	80	100	80
Selected C_d (μm)	60	60	60	60	95	120	95
Irradiance ($\mu\text{W}/\text{cm}^2$)	2.03	2.03	2.03	2.03	2.03	2.03	2.03
Calculated V_s ($\mu\text{m}/\text{s}$)	639.2	639.2	639.2	639.2	610.2	590.3	610.2
Applied V_s ($\mu\text{m}/\text{s}$)	640	640	640	640	610	590	610
Part dimension (mm^3)	$20 \times 7 \times 2$	$20 \times 7 \times 6$	$20 \times 7 \times 2$	$20 \times 7 \times 6$	$22 \times 7 \times 5$	$24 \times 7 \times 40$	$22 \times 8 \times 37$
Total built time (min)	38	114	38	114	65	374	390
Pattern width, N_p (pixel)	360	360	360	360	400	440	400
Frame width, N_f (pixel)	150	150	150	150	150	150	150
Pixel pitch, ρ (μm)	56	56	56	56	56	56	56

Table 4

Summary of feature size for various parts.

Model (Fig. 11)	Dimension	Desired (mm)	Data measured (mm)						Average (mm)	Standard deviation (μm)
(a)	X	3.696	3.90	3.80	3.90	3.95	3.85	3.88	13	
	Y	3.696	3.95	3.90	3.85	4.05	3.90	3.93	23	
(c)	X	3.584	3.80	3.75	3.95	3.85	3.95	3.86	32	
	Y	3.584	3.65	3.55	3.65	3.70	3.60	3.63	13	
(e)	X	4.256	4.50	4.35	4.55	4.40	4.40	4.44	27	
	Y	4.256	4.55	4.45	4.40	4.45	4.55	4.48	18	
(g)	X	4.144	4.45	4.65	4.60	4.60	4.60	4.58	23	
	Y	4.144	4.10	4.25	4.30	4.25	4.15	4.21	27	

It should be pointed out that the overhanging part is created without any support. In addition, any mismatch between the layer thickness and the illuminated energy causes over/under exposure.

7. Summary and conclusion

Dynamic (maskless) PSL systems are increasingly adopted by 3D printing industry to meet the ever increasing demands of rapid fabrication. A novel digital technique for the production of large parts with small feature sizes has been described here using commercially available DMDs. In this work, the feasibility, capability and versatility of the proposed idea as well as repeatability of the process is demonstrated.

The system is mathematically modeled and practically implemented. The system has a simple inexpensive structure. Five different CAD models are designed and manufactured. Each model is built five times and measured. The statistical data shows that the standard deviation is lower than 32 μm which means the repeatability based on 2σ criteria is less than 64 μm. Additionally, two complex parts with overhanging and freeform geometries are produced to show the capability and versatility of SPSL for real industrial applications.

The authors believe SPSL is capable of manufacturing large/micro parts with low-cost machines while maintaining the resolution. They also expect the non-uniformity of irradiance over DMD to be eliminated. These will be the concept of our next publications.

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Appendix.

To show how V_s is calculated in Table 3, consider case (a). In order to achieve a layer with the thickness of 40 μm, C_d should be larger than 40 μm (for example 60 μm here) to ensure bonding between layers. For $I_{\max} = 2.06 \text{ mW/cm}^2$, $D_p = 754 \text{ μm}$ and $E_c = 25 \text{ mJ/cm}^2$ (mentioned in Section 4) as well as $\rho = 56 \text{ μm}$, $N_f = 150$ pixels and $C_d = 60 \text{ μm}$ (Table 3), the value of V_s based on Eq. (6) is:

$$V_s = I_{\max} \cdot \rho \cdot \frac{N_f}{E_c} \cdot \exp\left(\frac{-C_d}{D_p}\right)$$

$$V_s = \frac{2.06 (\text{mW/cm}^2) \times 56 (\text{μm}) \times 150}{25 (\text{mJ/cm}^2)} \cdot \exp\left(-\frac{60 (\text{μm})}{754 (\text{μm})}\right) = 639.2 (\text{μm/s})$$

For some motion controller like the one used in the current research, the calculated value of V_s should be rounded down/up to an eligible value. Here $V_s = 640 \text{ μm/s}$ is replaced instead.

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